STUDENT LEAVE MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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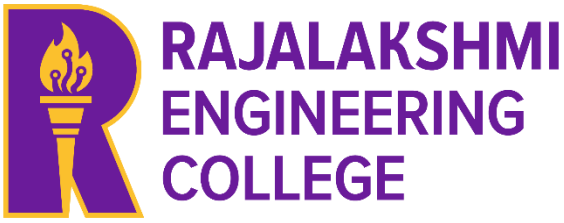
in partial fulfilment of the End Semester Examination for the course

CB23F35 – LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING

COMPUTER SCIENCE AND ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

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RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "Student Leave Management System" is the bonafide work of SOAMESHWARAN D (240701520), who carried out the project work using the Zoho Creator CB23F35- low-code application development platform during the academic year [2026-2027], Semester[V] under my supervision.

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DECLARATION

I, SOAMESHWARAN D (240701520), hereby declare that the project report titled “Student Leave Management System”, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai

Date:

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ABSTRACT

The Student Leave Management System is a low-code application developed using Zoho Creator to digitise and simplify the student leave application and approval process. In a conventional process, students submit leave requests through paper forms or informal messages. Faculty members review requests manually, and administrators maintain leave details in registers or spreadsheets. This approach can lead to delays, missing records, difficulty tracking application status and repeated manual work.

The proposed system provides a centralised platform in which student details are maintained and leave applications are submitted with the leave dates, leave type, reason and supporting document, where applicable. The application validates the entered dates, calculates the total number of leave days and assigns a Pending status to a newly submitted request. Faculty members can review the request and approve or reject it with remarks. Email notifications can be sent to the concerned faculty member when a request is submitted and to the student when the status is updated.

The application is organised around student registration, faculty registration and leave application records. Reports help users view pending, approved and rejected requests, while an administrator dashboard provides an overview of leave activity. Role-based access is intended to separate the activities of students, faculty members and administrators. The system aims to reduce paperwork, improve the accuracy of leave records, provide better visibility of request status and make the leave approval process more efficient.

Keywords: Student Leave Management; Zoho Creator; Low-Code Development; Deluge Scripting; Leave Approval; Workflow Automation.

PROJECT AT A GLANCE

Attribute	Details
Project Title	Student Leave Management System
Application Domain	Education – student administration
Platform	Zoho Creator
Application Type	Low-code web application
Main Users	Student, Faculty, Administrator
Main Forms	Student Registration; Faculty Registration; Leave Application
Core Process	Submit leave request → Faculty review → Approve / Reject → Notify student
Automation	Date validation, leave-day calculation, default Pending status and email notifications
Reports	Pending, Approved and Rejected leave applications
Dashboard	Summary of leave application status and counts
Project Status	Update this row to match the features completed in your application
Live Application URL	[Paste your published Zoho Creator URL here]

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
FR	Functional Requirement
NFR	Non-Functional Requirement
RBAC	Role-Based Access Control
UI	User Interface
UAT	User Acceptance Testing
LCAP	Low-Code Application Platform
IST	Indian Standard Time

CHAPTER 1

INTRODUCTION

1.1 Background

Educational institutions handle student leave requests throughout the academic year. Students may need leave for personal reasons, illness, family commitments, participation in events or other valid circumstances. In a manual process, a student writes a request, obtains the required approval and waits for the decision. Faculty members may need to check dates, student details and supporting documents before making a decision. Administrative staff may then update registers or spreadsheets.

The Student Leave Management System is designed to bring these activities into a single digital application. The project uses Zoho Creator, a low-code platform that provides visual form creation, reports, workflows and Deluge scripting. Students can submit leave details, faculty members can review and update the request status, and administrators can monitor records.

1.2 Problem Statement

The traditional leave process depends on paper forms, direct messages or separate spreadsheets. As a result, applications may be misplaced, approval status may not be communicated promptly and staff may spend time checking dates and updating records manually. There may also be difficulty obtaining a consolidated view of pending, approved and rejected applications. The problem addressed by this project is to develop a centralised and easy-to-use application that records leave requests, supports faculty approval and provides clear status tracking.

- Paper-based requests can be misplaced or damaged.
- Students may not know whether their application is pending, approved or rejected.
- Faculty members must manually review dates and calculate the leave duration.
- Maintaining and searching historical leave records can take additional time.
- Separate communication channels can result in delayed or missed updates.

1.3 Objectives

- To create and maintain student and faculty details in a centralised application.

- To allow students to submit leave applications with start date, end date, leave type, reason and supporting document, where required.
- To validate leave dates and calculate the total number of leave days automatically.
- To route submitted applications to the appropriate faculty member for review.
- To allow faculty members to approve or reject requests and enter remarks.
- To notify faculty members about new applications and students about status changes through email, where configured.
- To provide reports and a dashboard for monitoring leave applications.
- To apply role-based access so that students, faculty and administrators use the relevant application functions.

1.4 Scope of the Project

The project covers student and faculty record maintenance, leave application submission, date validation, leave-day calculation, request status management, faculty review, remarks, email notifications, reports and an administrator dashboard. Supporting documents may be attached to a leave request if the configured form includes an upload field.

The exact scope depends on the features implemented in the final Zoho Creator application. Features such as attendance-system integration, mobile push notifications, leave balance calculation, multi-level approval and integration with a college ERP may be considered future enhancements unless they have been completed and tested.

1.5 Justification for Low-Code Development

Zoho Creator provides visual form builders, report components, workflow configuration and Deluge scripting. These features are suitable for a process-oriented application such as leave management, where the main requirements involve collecting information, validating records, routing requests and notifying users. A low-code approach reduces the amount of code required and makes it easier to modify form fields and workflows during development.

1.6 Organisation of the Report

Chapter 1 introduces the project. Chapter 2 studies the existing process and proposed solution. Chapter 3 describes the requirements. Chapter 4 explains the system design, data model and workflow. Chapter 5 presents the implementation. Chapter 6 describes testing and debugging. Chapter 7 discusses security

and deployment. Chapter 8 presents the results and limitations. Chapter 9 concludes the report and lists possible future enhancements.

CHAPTER 2

LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In a conventional college environment, a student may obtain a leave form, enter personal and leave details, attach supporting documents and submit the request to a class advisor or faculty member. The faculty member checks the request and communicates approval or rejection. A staff member may maintain the final details in a register or spreadsheet.

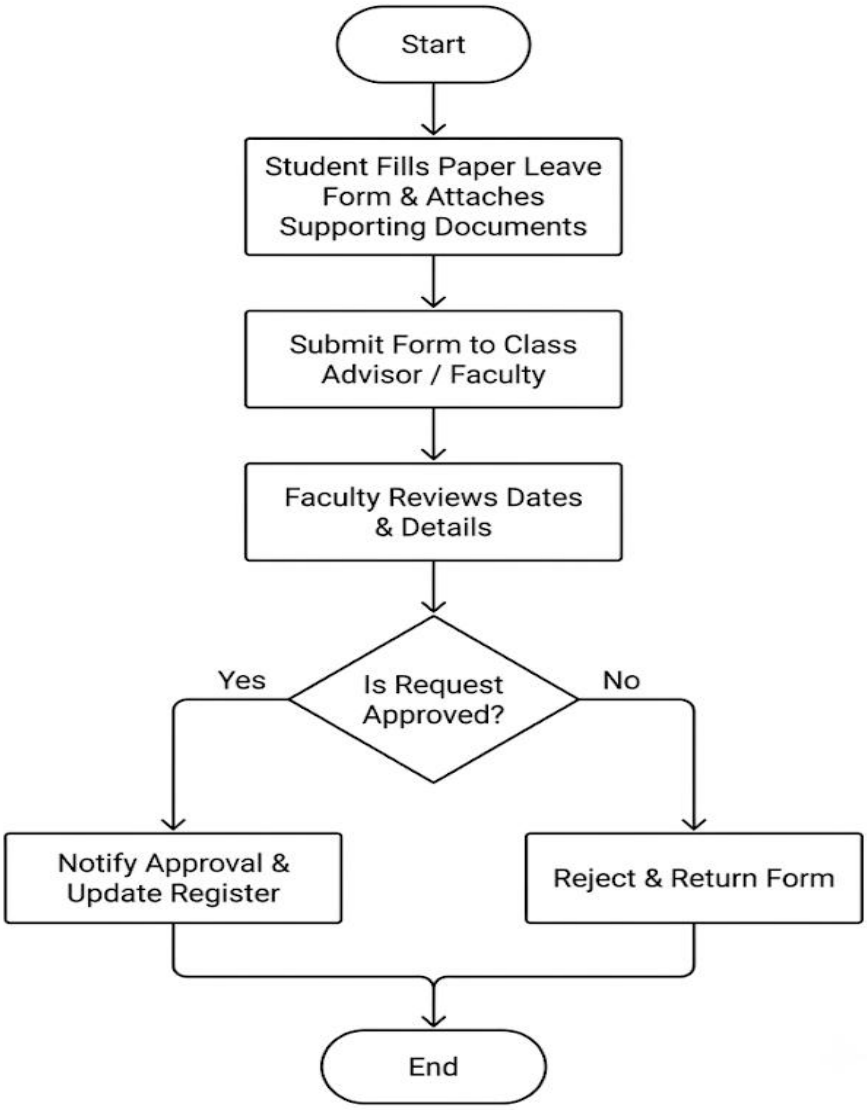


Figure 2.1: Existing Manual Leave Application Process

2.2 Limitations of the Existing System

- Manual entry increases the possibility of incomplete or incorrect information.
- Students may need to contact faculty members to know the status of an application.
- Date calculations and record updates require manual effort.
- Searching for old applications can be time-consuming.
- There may be no central dashboard showing the number of pending, approved and rejected requests.
- Documents and messages may be stored in different locations.

2.3 Proposed System

The proposed Student Leave Management System stores student details, faculty details and leave applications in a Zoho Creator application. The student submits a request through the Leave Application form. The application validates the date range, calculates the number of leave days and sets the initial status to Pending. The designated faculty member reviews the request and records an approval decision and remarks. The system can send email notifications at submission and when the request status changes. Reports and a dashboard provide a consolidated view of leave records.

2.4 Literature / Platform Study

The platform study focuses on the low-code capabilities used in this project: data forms, lookup relationships, validation rules, Deluge scripts, workflows, reports, dashboards, file-upload fields and user permissions. The official Zoho Creator documentation should be consulted for the exact behaviour and availability of each feature in the account edition used.

Platform feature	Use in this project
Forms	Capture student, faculty and leave application information.
Lookup fields	Connect leave requests with student and faculty records.
Deluge scripting	Validate dates, calculate leave days and set default values.
Workflows	Trigger notifications and automate status-related actions.

Reports	Display leave requests by status or other selected criteria.
Pages / Dashboard	Summarise application activity for administrators.
Permissions	Restrict access according to user role.

CHAPTER 3

REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Role	Description	Responsibilities
Student	Person applying for leave.	Enter leave dates, type and reason; attach document if required; view own request status.
Faculty / Class Advisor	Reviews leave applications assigned to them.	Check request details; approve or reject; enter remarks; monitor assigned requests.
Administrator	Maintains and monitors the application.	Manage master records and access; view reports and dashboard; support application maintenance.

3.2 Functional Requirements

ID	Functional requirement	Priority
FR-01	The system shall store student details such as name, register number, email, department, year and section.	High
FR-02	The system shall store faculty details and the faculty email / mapping required for routing requests.	High

FR-03	The system shall allow a student to submit a leave application.	High
FR-04	The leave application shall capture start date, end date, leave type and reason.	High
FR-05	The system shall allow a supporting document to be uploaded when the form is configured for attachments.	Medium
FR-06	The system shall validate that the leave dates are present and the end date is not earlier than the start date.	High
FR-07	The system shall calculate the total leave days from the selected date range.	High
FR-08	A newly submitted application shall be assigned Pending status.	High
FR-09	Faculty shall be able to approve or reject a leave request and enter remarks.	High
FR-10	The system shall notify the faculty member when a new request is submitted, where email workflow is enabled.	Medium
FR-11	The system shall notify the student when the application	Medium

	status is updated, where email workflow is enabled.	
FR-12	The system shall provide reports for Pending, Approved and Rejected requests.	Medium
FR-13	The administrator dashboard shall summarise leave application counts.	Medium
FR-14	The system shall provide role-based access to relevant forms and reports.	High

3.3 Non-Functional Requirements

Category	Requirement
Usability	Forms should use clear labels and simple instructions.
Reliability	Submitted records and approval decisions should be stored consistently.
Security	Users should access only the records and actions permitted to their role.
Performance	Forms and reports should load within a reasonable time under normal usage.
Maintainability	Administrators should be able to update fields, reports and workflows when requirements change.
Availability	The application should be accessible when the Zoho Creator service and internet connection are available.

Privacy	Student contact details and uploaded documents should be protected from unauthorised access.

3.4 Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (edition used: [enter edition])
Scripting	Deluge
Browser	Recent version of Google Chrome, Microsoft Edge or Mozilla Firefox
Operating system	Windows / macOS / Linux / Android / iOS supported browser or app
Hardware	Computer or smartphone with internet access
Other services	Email service / Zoho Mail, if configured for notifications

3.5 Use Case Description

Primary use case: Submit and process a student leave application.

Item	Description
Use Case ID	UC-01
Use Case Name	Submit Leave Application
Actors	Student; Faculty; Administrator
Pre-condition	Student record exists and the student can access the application.
Main flow	1. Student opens the Leave Application form. 2. Student enters leave dates, type and reason. 3. Student attaches a supporting document if required. 4. Student submits the form. 5. System

	validates the date range and calculates leave days. 6. System saves the request with Pending status. 7. Faculty member is notified if email notification is configured.
Alternate flow	If required dates are missing or the end date is earlier than the start date, the system displays a validation message and prevents submission.
Post-condition	A valid leave request is stored and can be reviewed by the assigned faculty member.

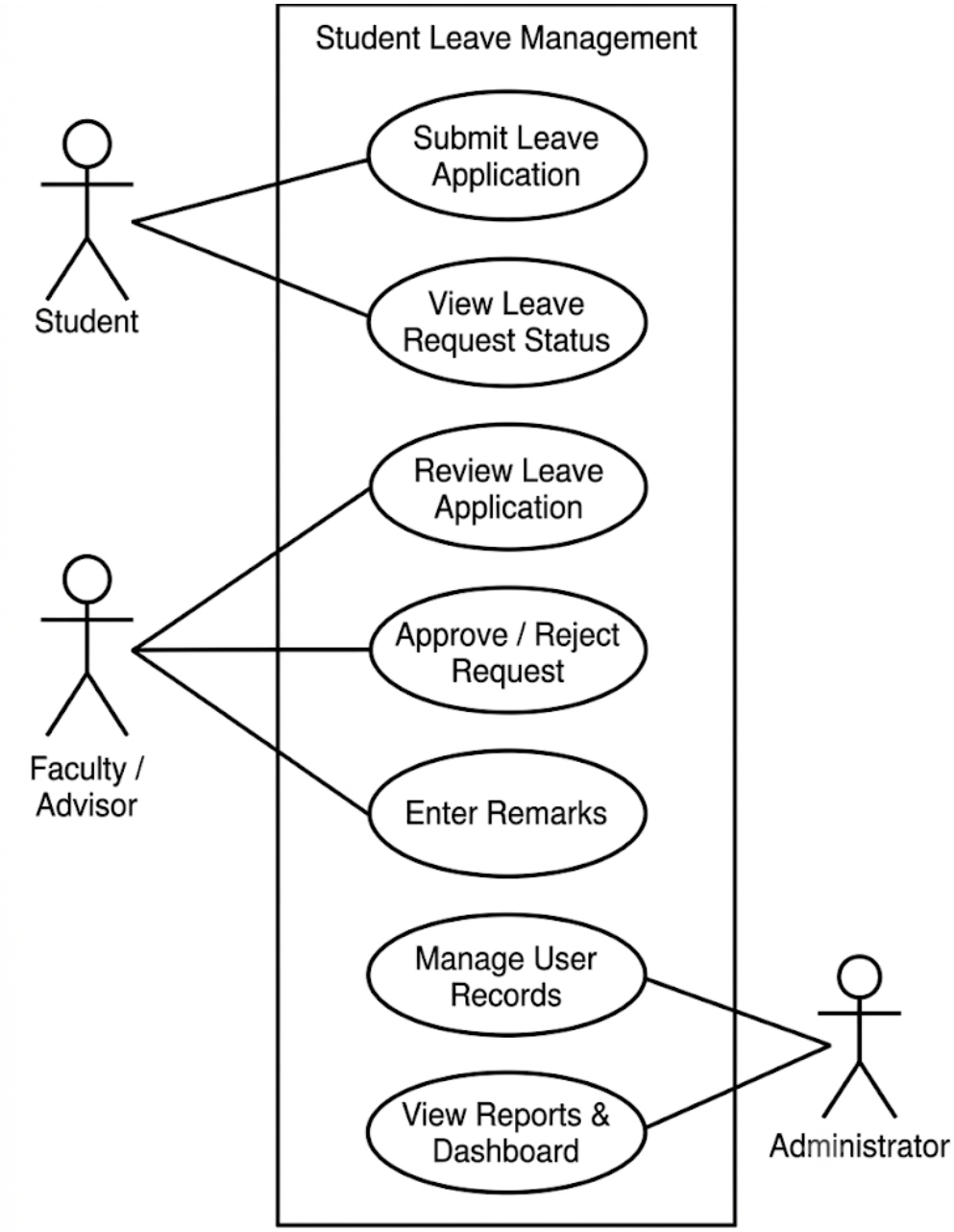


Figure 3.1: Use Case Diagram

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

The proposed application can be represented using four logical layers. The presentation layer contains the forms, reports and dashboard used by students, faculty and administrators. The application layer contains Zoho Creator workflows and Deluge scripts. The data layer stores student, faculty and leave application records. The notification layer sends email messages when the configured workflows are triggered.

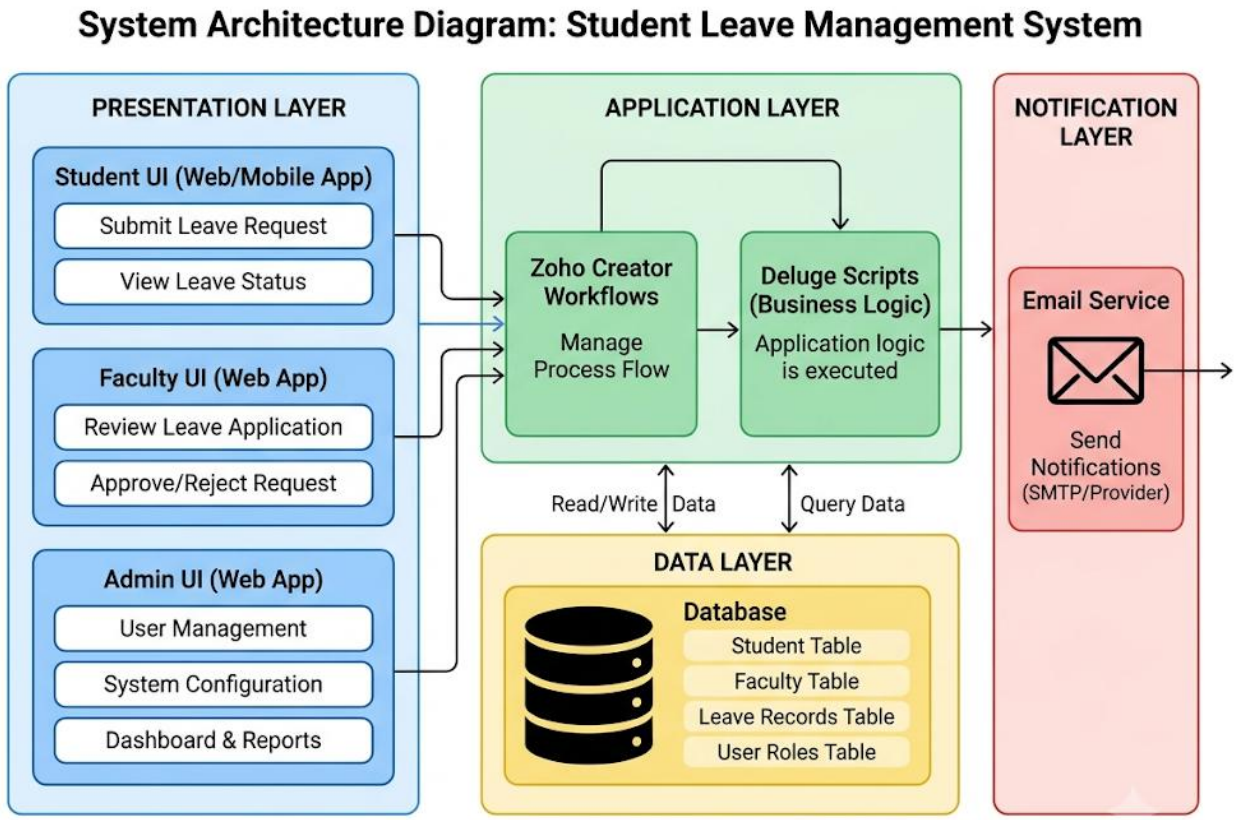


Figure 2.1: Presentation Layer: Student Leaver UI (Web InieroSystem) Layer: Zoho Creator Workflows, Manage Student Leave, Application Layer Applicationy UI (Web App) I, Review ʁ/ Process Logic anide Scripts (System Configuration), Application Logic, (Deluge Management, User Roles Configuration, Bærige Configurach, Dashboard & Reports.

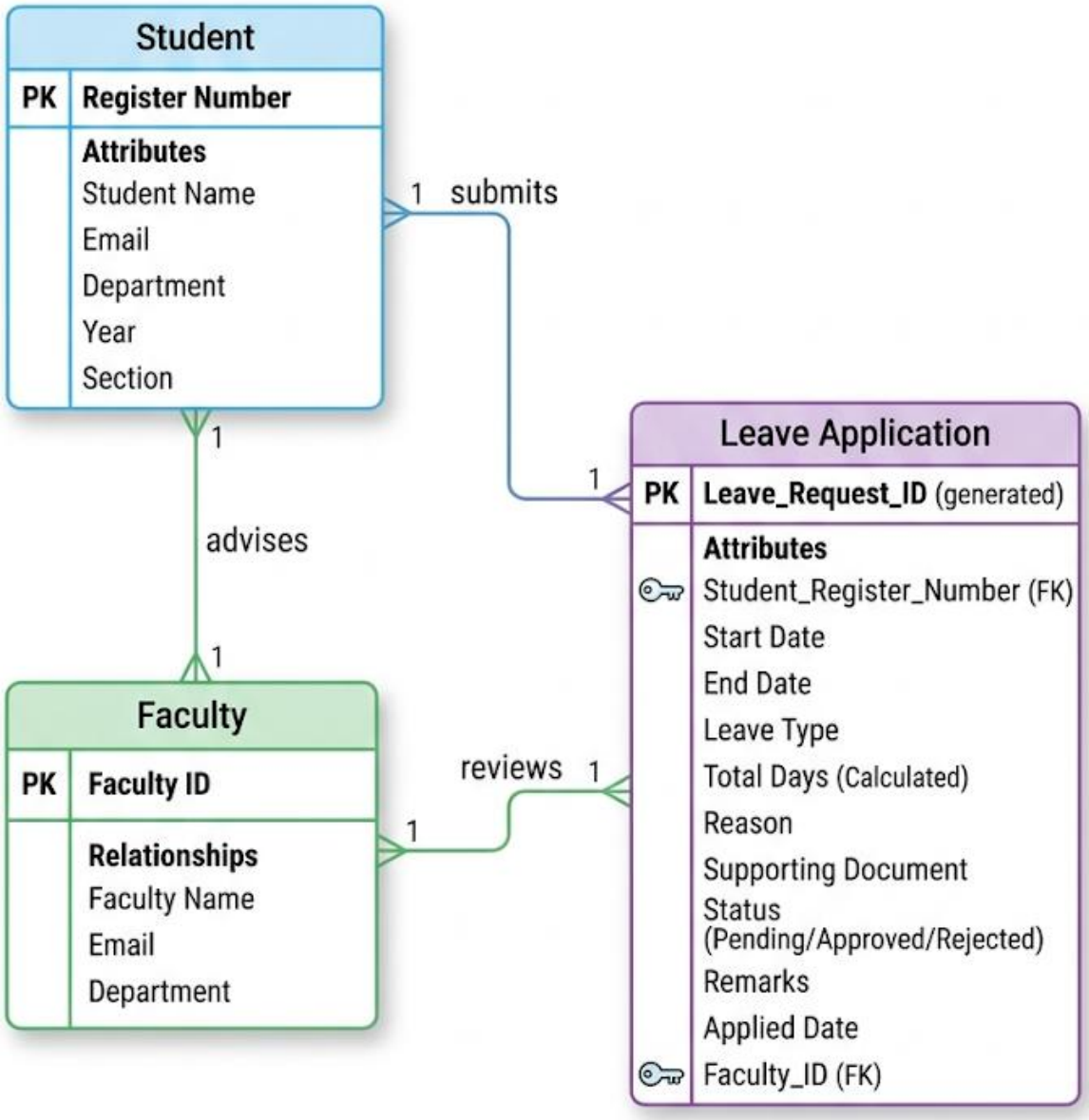
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4.2 Data Model

The main data entities are Student, Faculty and Leave Application. A student can submit multiple leave applications. A faculty member may be associated with or assigned multiple leave applications. The Leave Application record acts as the transaction record and contains lookup fields that connect it to the relevant student and faculty records.

Entity / Form	Example fields	Relationship
Student_Registration	Student Name, Register Number, Email, Department, Year, Section, Faculty Email	One student can have many leave applications.
Faculty_Registration	Faculty Name, Faculty ID, Email, Department	One faculty member can review multiple applications.
Leave_Application	Student, Start Date, End Date, Leave Type, Total Days, Reason, Supporting Document, Faculty Email, Status, Remarks, Applied Date	Each application belongs to a student and is routed to a faculty member.

Figure 4.2: Entity-Relationship Diagram: Student Leave Management System



Entity-Relationship Dıə Eſammounted Student Leave Management PKs; **PK**: Student Leave Management System desin **Student**, 3, **Faculty**, **Leave Application** can number : I", **Faculty**', reviews, '1, 'conertI ſen on, Remarkı reg., 'A, Faculty member is assigned **is assigned to multiple students** for leave routing.

4.3 Form Design

4.3.1 Student Registration Form

The Student Registration form stores the identity and academic details needed to identify the applicant and route the leave request. The register number should be treated as a unique identifier if uniqueness validation has been configured.

Field	Suggested field type	Purpose
Student Name	Name / Single Line	Identifies the student.
Register Number	Single Line	Unique student identifier.
Email	Email	Contact and notification address.
Department	Dropdown	Academic department.
Year	Dropdown / Number	Current year of study.
Section	Dropdown / Single Line	Class section.
Faculty Email	Email / Lookup	Faculty routing information.

4.3.2 Faculty Registration Form

The Faculty Registration form stores faculty details used for identifying the reviewer and routing applications.

Field	Suggested field type	Purpose
Faculty Name	Name / Single Line	Faculty member name.
Faculty ID	Single Line	Faculty identifier.
Email	Email	Notification address.
Department	Dropdown	Department association.

4.3.3 Leave Application Form

The Leave Application form captures the leave request and its processing status. The exact field names and types should be updated to match the form created in Zoho Creator.

Field	Suggested field type	Purpose
Student	Lookup	Selects the student record.
Start Date	Date	First day of leave.
End Date	Date	Last day of leave.
Leave Type	Dropdown	Type or category of leave.
Total Days	Number	Automatically calculated duration.
Reason	Multi Line	Student's explanation.
Supporting Document	File Upload	Supporting proof, if required.
Faculty Email / Faculty	Email / Lookup	Identifies the reviewer.
Status	Dropdown	Pending, Approved or Rejected.
Remarks	Multi Line	Faculty decision comments.
Applied Date	Date-Time	Date and time of submission.

4.4 Workflow Design

The normal workflow begins when a student submits a leave request. The system validates the date fields and checks that the end date is not earlier than the start date. It then calculates the number of days using the configured date calculation, assigns Pending status and saves the request. A notification is sent to the faculty member if the email workflow is enabled. The faculty member reviews the request and changes the status to Approved or Rejected, optionally entering remarks. A status-change workflow can then notify the student.

FIGURE: STUDENT LEAVE APPROVAL PROCESS FLOWCHART



Figure: Leave Approval Workflow.

4.5 User Interface and Navigation Design

The application navigation should provide clear access to student registration, faculty registration, leave application entry, faculty review, leave reports and the administrator dashboard. Menu labels should be consistent with the names used in the published application.

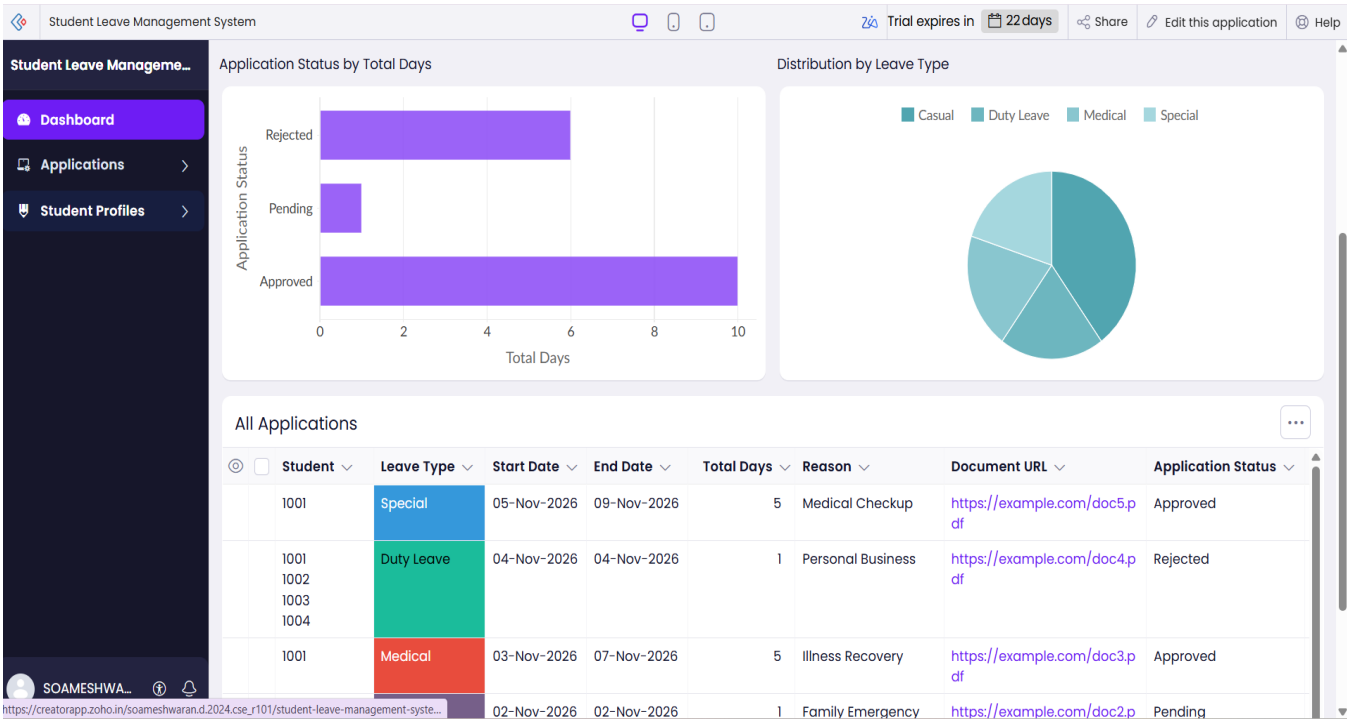


Figure: Application Navigation / Home Page.

4.6 Access Control Design

Role	Expected access
Student	Submit leave applications and view their own applications, subject to configured permissions.
Faculty	View assigned requests and update decision fields such as status and remarks.

Administrator	Manage master records, view all applications, access reports and dashboard, and maintain the application.

CHAPTER 5

IMPLEMENTATION

5.1 Application Setup

The application was created in Zoho Creator. The app name, time zone, date format, menu structure and user permissions should be documented according to the actual configuration. The forms and workflows were developed using the visual builders, with Deluge scripts used for the validation and automation logic.

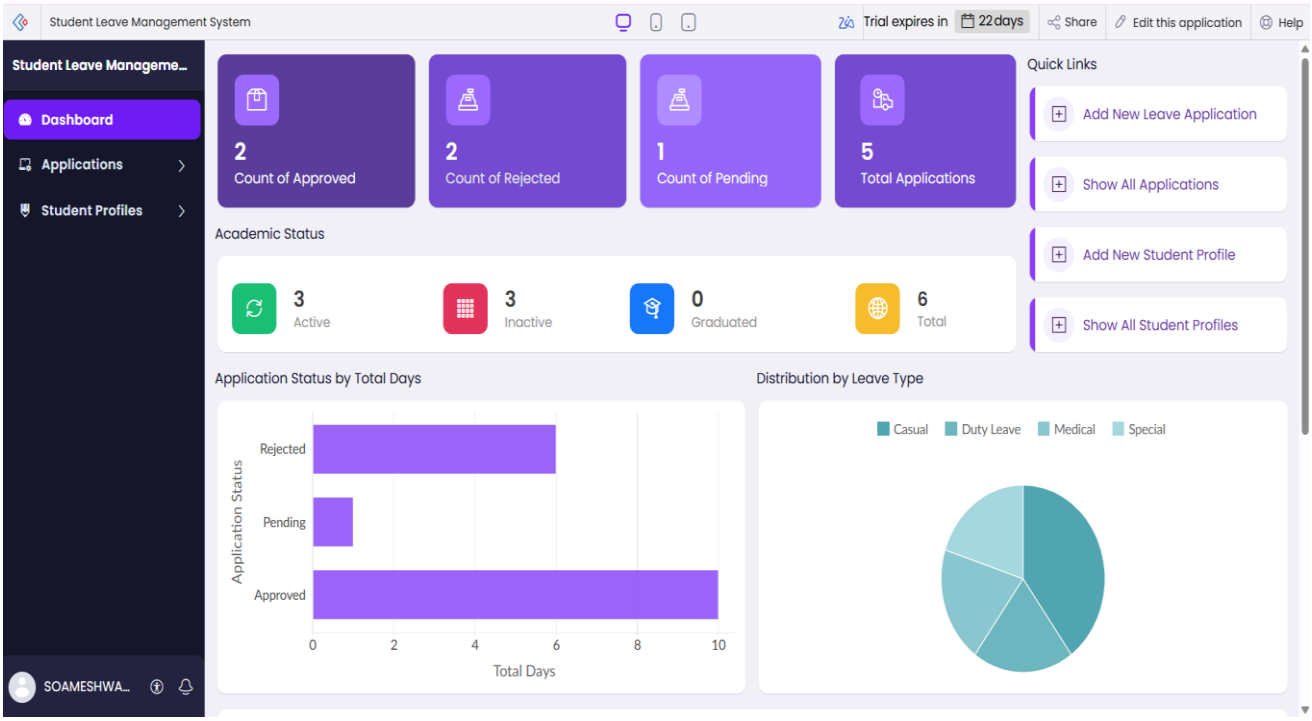


Figure: Zoho Creator Application Home.

5.2 Student Registration Form

The Student Registration form is used to create and maintain student records. The register number, email and academic details provide a reference for each student. If the application uses a unique register number, duplicate values should be prevented through a validation rule.

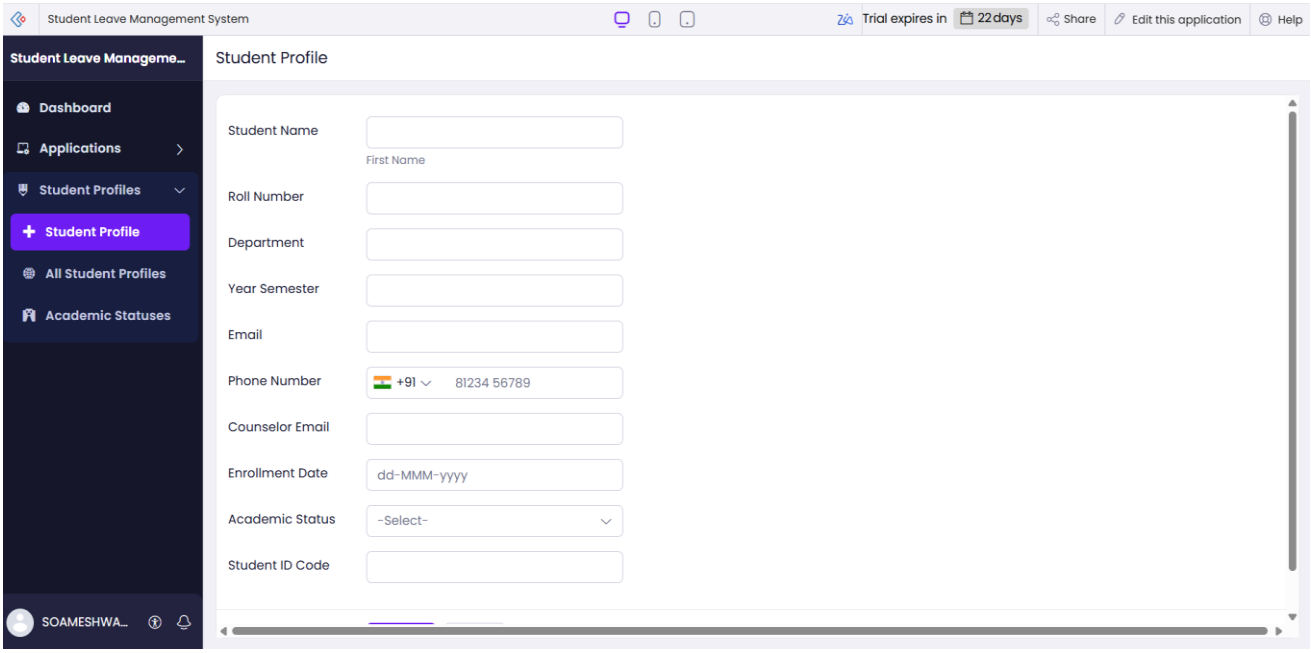


Figure: Student Registration – Live Form.

5.3 Leave Application Form

Students use the Leave Application form to enter the leave period, leave type and reason. A supporting document can be attached if the file-upload field is part of the implemented form. The application stores the calculated number of leave days and the current approval status.

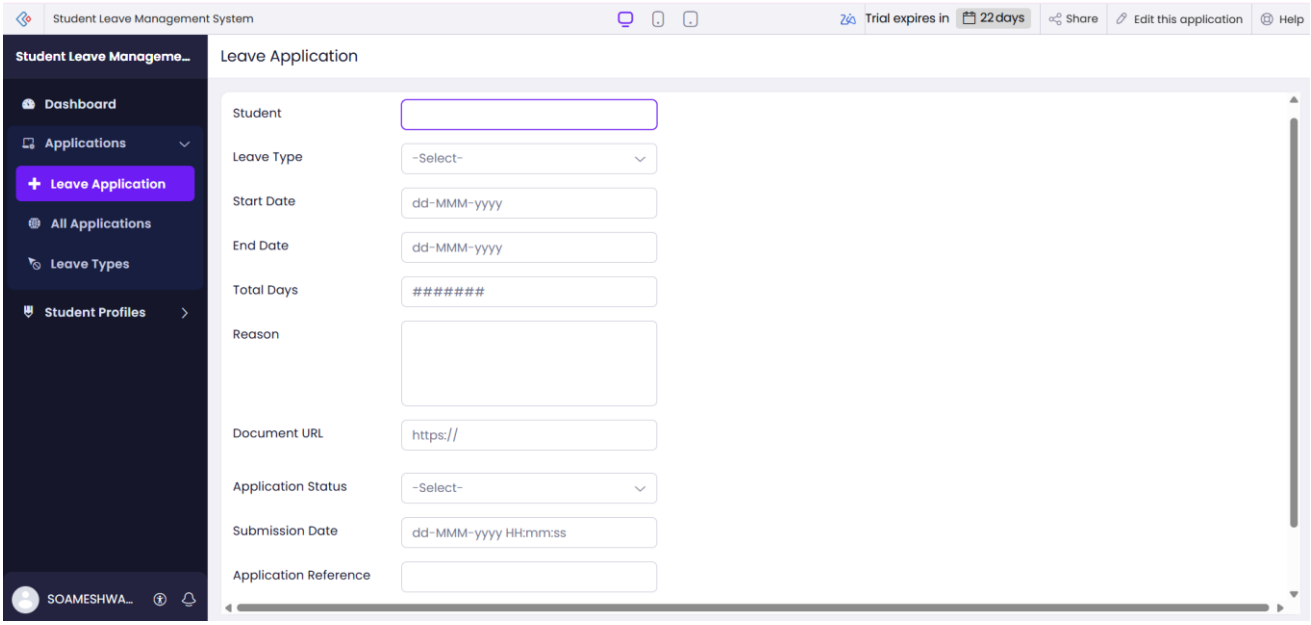


Figure: Leave Application – Student View.

5.4 Deluge Scripting

Deluge is used to implement the application-specific logic. The following table describes the intended scripts based on the project workflow. Replace the script names and code with the exact version used in your application.

Script / Event	Purpose	Logic
Date validation – On Validate	Prevent invalid date ranges.	Check that Start Date and End Date are entered and End Date is not earlier than Start Date.
Leave-day calculation	Calculate total leave days.	Calculate the difference between the dates and add one if both start and end dates are counted.
Default status – On Validate / On Success	Set initial status.	Assign Pending to a newly submitted request.
Submission notification	Notify faculty.	Send an email containing student name, leave period, leave type and reason, if configured.
Status-change notification	Notify student.	Send an email containing the decision and faculty remarks when status changes.

Example calculation logic (adapt field link names and syntax to the actual form):

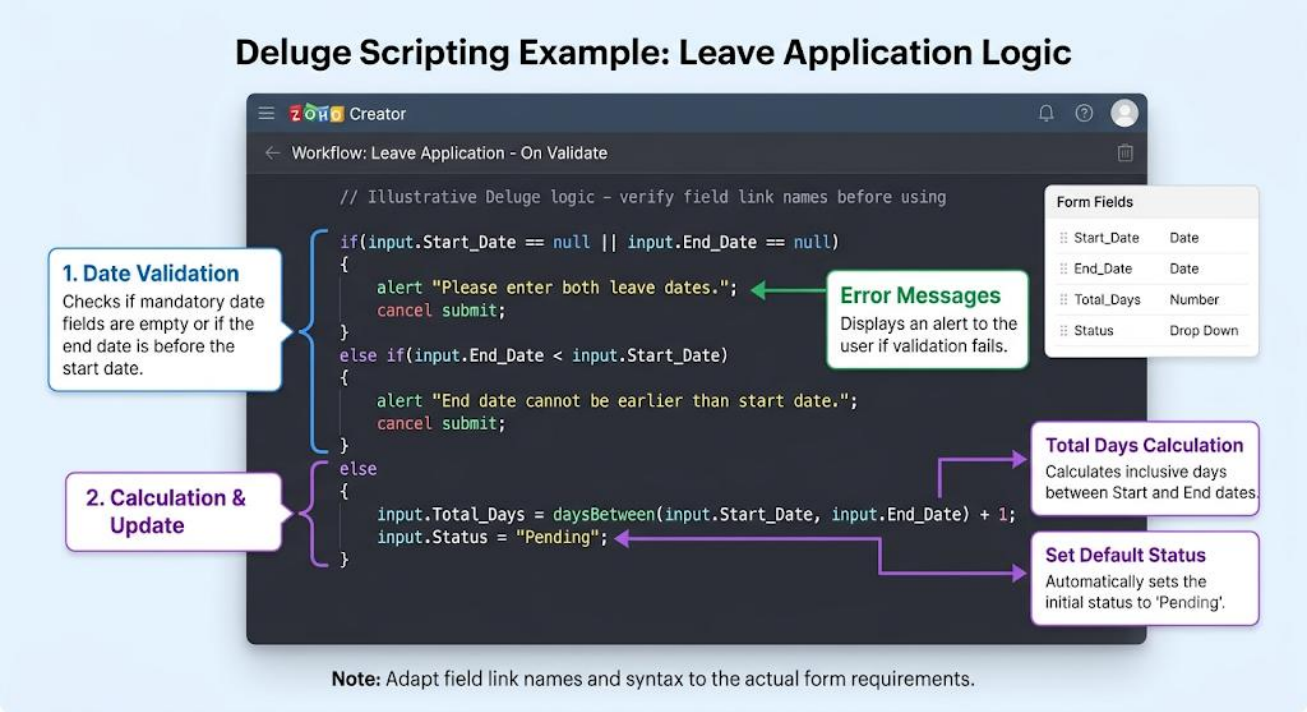


Figure: Deluge Script / Workflow Editor.

5.5 Workflows and Notifications

Workflow	Trigger	Action
New Leave Request	After successful submission	Notify the faculty member assigned to the request, if configured.
Leave Status Updated	After status edit	Notify the student about approval or rejection, including remarks if configured.
Date Validation	Before saving the request	Prevent missing or invalid date ranges.
Leave Days Calculation	During form validation / submission	Calculate and store the number of leave days.

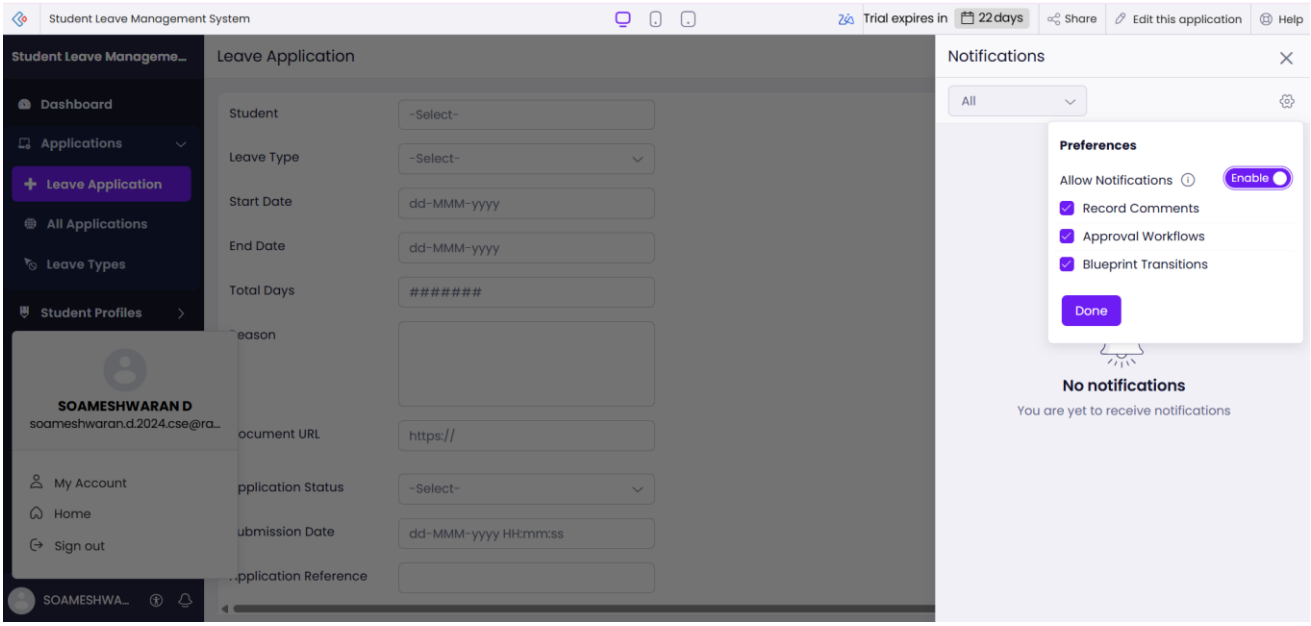


Figure: Email Notification Evidence.

5.6 Reports and Dashboard

Reports provide filtered views of leave applications. Typical report views include all leave applications, Pending applications, Approved applications and Rejected applications. The administrator dashboard can show the total number of applications and status-wise counts or charts, depending on the components configured.

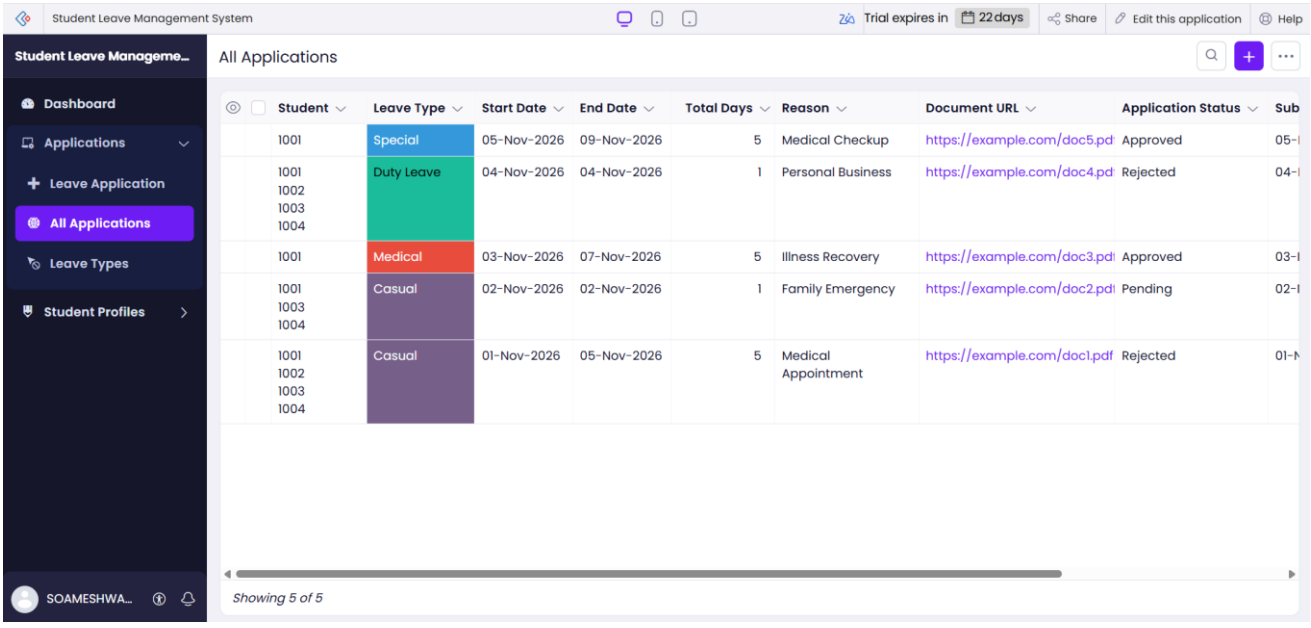


Figure: Pending Leave Applications Report.

CHAPTER 6

TESTING AND DEBUGGING

6.1 Testing Strategy

Testing is performed to confirm that forms accept valid information, validation rules reject invalid information, leave-day calculations are correct, notifications are triggered as expected and users can access only the functions permitted to their roles. Test with sample records and dummy contact details. Record the actual result and evidence after running each test.

6.2 Test Cases

Test ID	Test scenario	Expected result	Actual result / Status
TC-01	Submit a leave request with valid start and end dates.	Request is saved successfully with calculated total days and Pending status.	[Enter result]
TC-02	Leave start date is missing.	Validation message is displayed; request is not saved.	[Enter result]
TC-03	Leave end date is missing.	Validation message is displayed; request is not saved.	[Enter result]
TC-04	End date is earlier than start date.	Invalid date range is rejected.	[Enter result]
TC-05	Submit a request covering one day.	Total Days is calculated as 1 if both endpoints are counted.	[Enter result]
TC-06	Submit a request covering multiple days.	Total Days matches the selected date range and counting rule.	[Enter result]

TC-07	Faculty approves a Pending request.	Status changes to Approved and decision is stored.	[Enter result]
TC-08	Faculty rejects a Pending request and enters remarks.	Status changes to Rejected and remarks are stored.	[Enter result]
TC-09	Submit a request when notification workflow is enabled.	Assigned faculty receives the notification email.	[Enter result]
TC-10	Update request status when notification workflow is enabled.	Student receives the status notification email.	[Enter result]
TC-11	Student attempts to access another student's record.	Access is blocked or record is hidden according to configured permissions.	[Enter result]
TC-12	Open reports and dashboard as administrator.	Relevant leave records and summary values are displayed.	[Enter result]

6.3 Debugging Deluge Scripts

Deluge scripts should be tested with valid, invalid and boundary-value inputs. Use the Zoho Creator execution logs or workflow history to identify errors. Verify field link names, null checks, date formats, email addresses and workflow triggers. After correcting an error, repeat the related test case.

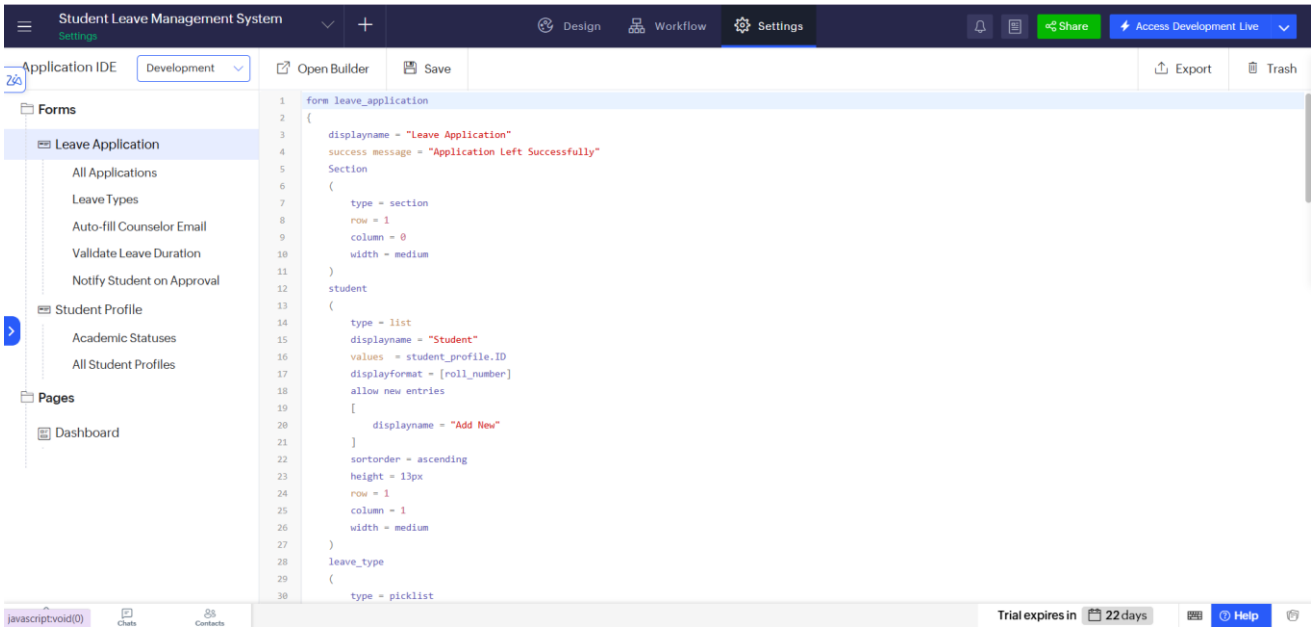


Figure: Deluge Execution Log / Debugging.

6.4 Defect Log

Defect ID	Issue identified	Correction made	Status
D-01	[Describe an issue found during testing]	[Describe the correction]	[Open / Closed]
D-02	[Describe an issue found during testing]	[Describe the correction]	[Open / Closed]
D-03	[Describe an issue found during testing]	[Describe the correction]	[Open / Closed]

6.5 User Acceptance Testing

User acceptance testing should be performed with representative student, faculty and administrator accounts. Confirm that each role can complete its expected tasks and that the leave approval process is understandable. Record the feedback received and any changes made before final submission.

User role	Task tested	Feedback / Observation
Student	Submit a leave request and check status.	[Enter feedback]
Faculty	Review and approve / reject a request.	[Enter feedback]
Administrator	View reports and dashboard.	[Enter feedback]

CHAPTER 7

SECURITY AND DEPLOYMENT

7.1 Deployment

The application should be tested in the development environment before it is published or shared with intended users. Before deployment, verify form validation, workflow triggers, user permissions, report filters, email notifications and the published application URL. Document the actual deployment method and date.

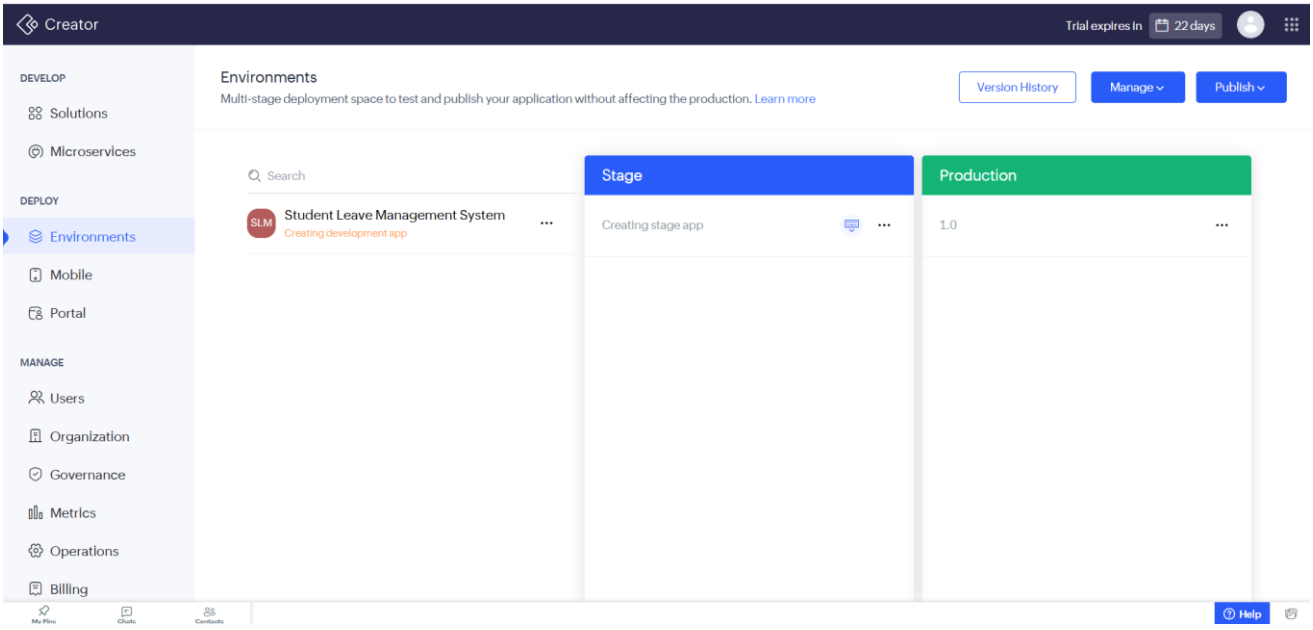


Figure: Publish / Deployment Screen.

7.2 Security and Privacy

- Use individual user accounts and avoid sharing administrator credentials.
- Restrict student access to their own leave applications wherever record-level permissions are available.
- Allow faculty members to view only the applications they are authorised to review.
- Limit administrator privileges to authorised staff.
- Protect uploaded medical or personal documents from public access.

- Use dummy data or mask names, register numbers, email addresses and documents in report screenshots.
- Review access permissions whenever a user changes role or leaves the institution.

7.3 Administration and Maintenance

The administrator should periodically verify student and faculty records, review pending requests, check notification delivery, inspect workflow errors and confirm that report filters continue to work. Changes to form fields or workflows should be tested before they are made available to all users.

CHAPTER 8

RESULTS AND DISCUSSION

8.1 Application Output

The completed application is expected to provide a digital path from leave request submission to faculty decision and student notification. The screenshots in this section should demonstrate the actual working application, not only the form-builder configuration.

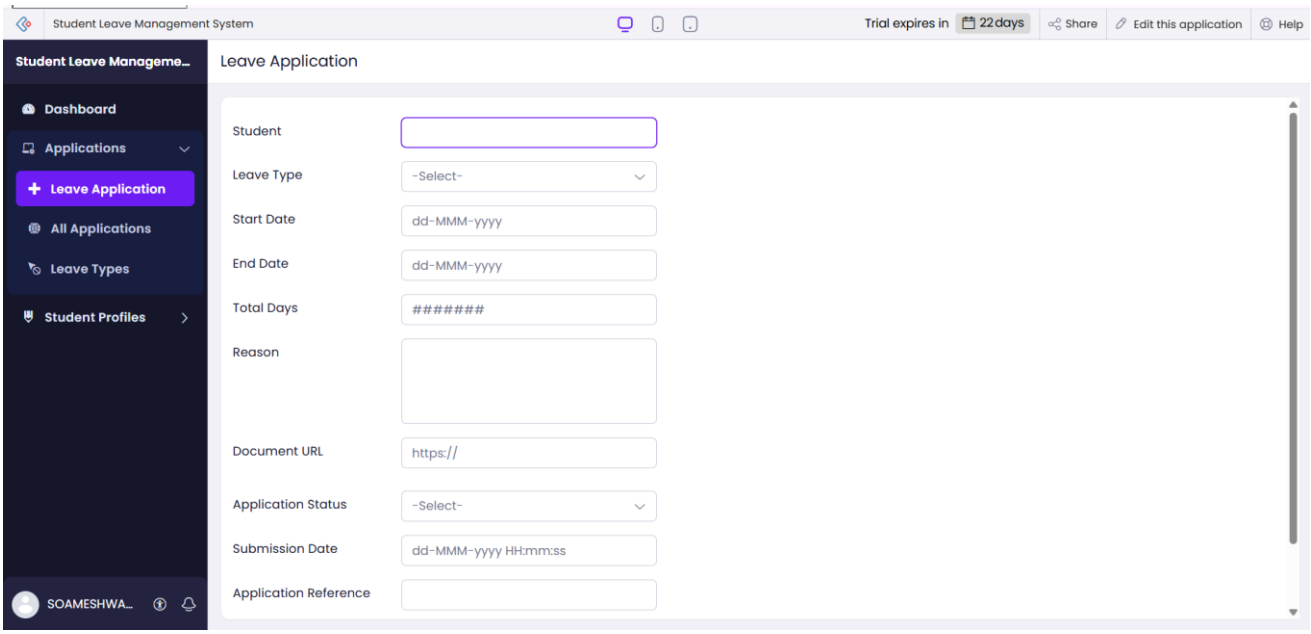


Figure: Student Submits Leave Request.

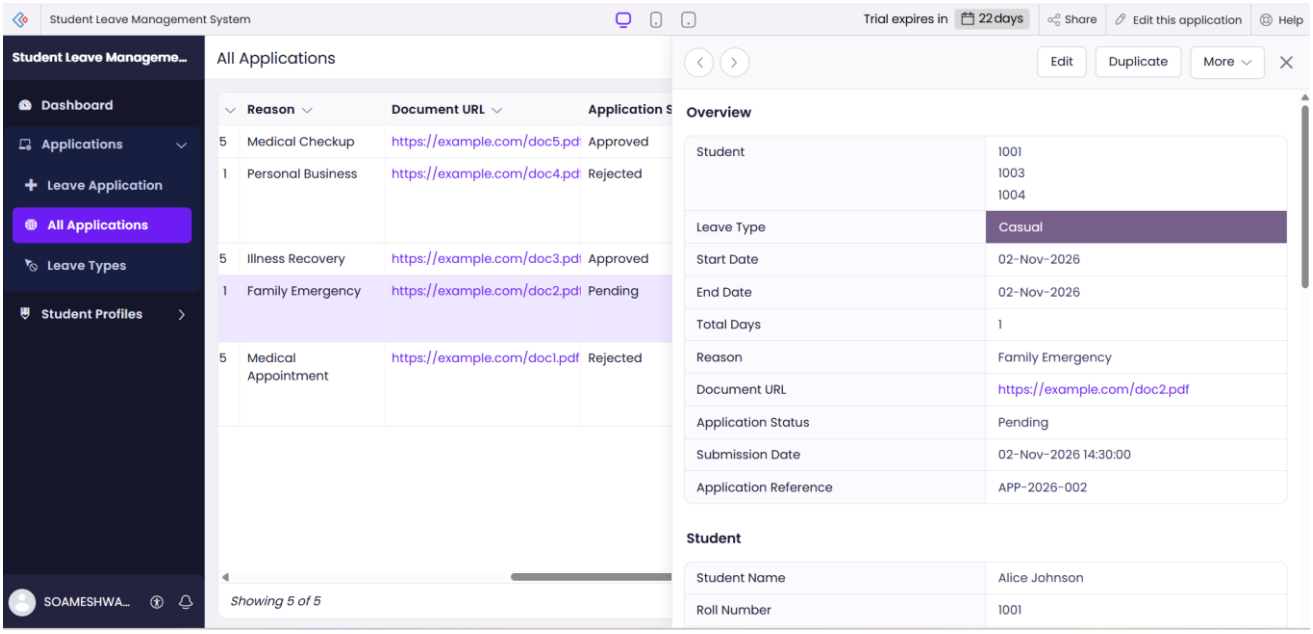


Figure: Faculty Reviews / Updates Request.

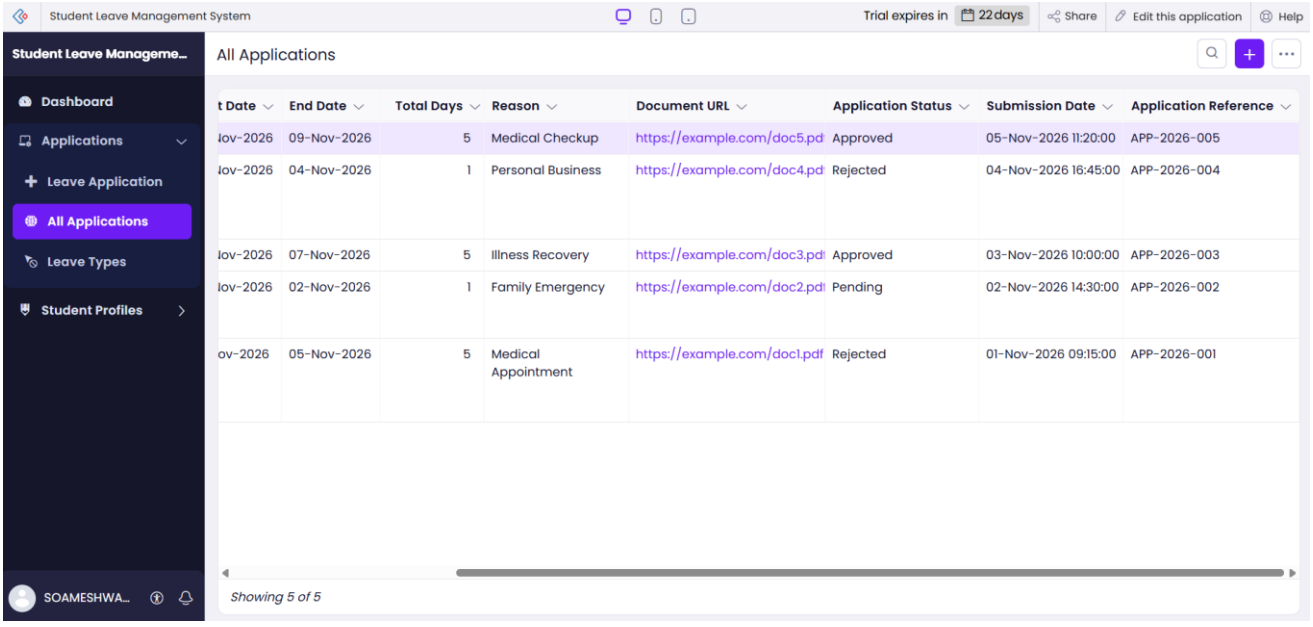


Figure: Student Views Final Status.

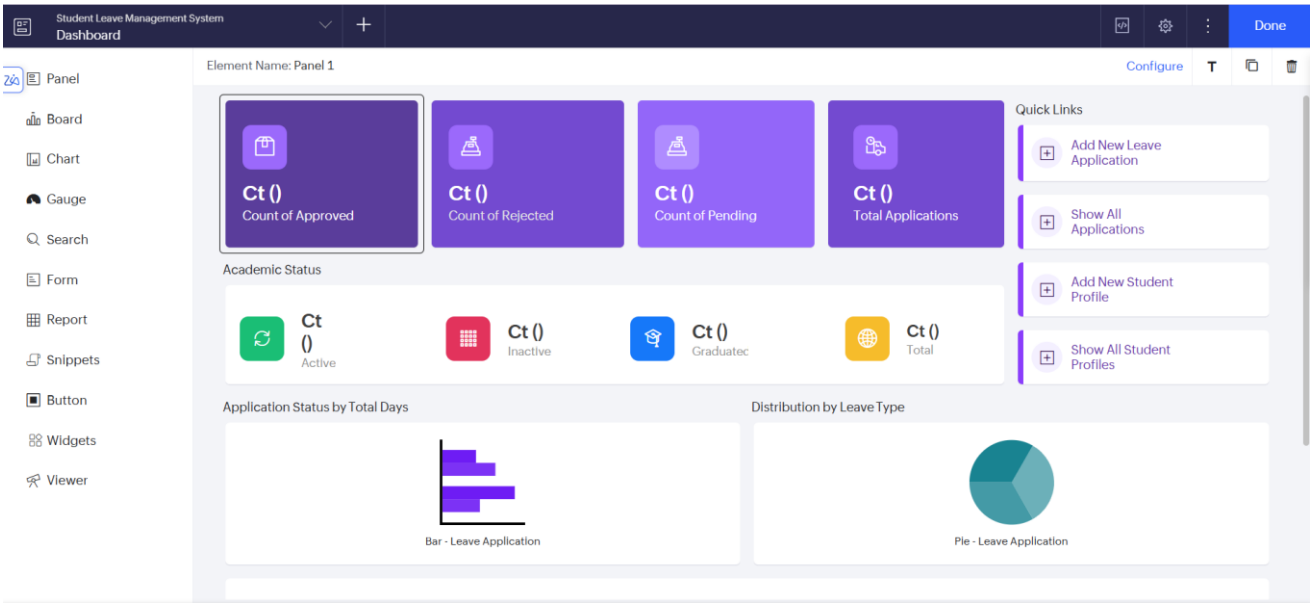


Figure: Dashboard Output.

8.2 Achievement of Objectives

Objective	Evidence to provide	Achievement
Centralised student and faculty records	Student and Faculty forms and reports	[Completed / Partial / Pending]
Online leave application	Leave Application form	[Completed / Partial / Pending]
Date validation and leave-day calculation	Deluge script and test cases	[Completed / Partial / Pending]
Faculty approval / rejection	Faculty review screen and status record	[Completed / Partial / Pending]
Email notifications	Sample notification email evidence	[Completed / Partial / Pending]
Reports and dashboard	Report and dashboard screenshots	[Completed / Partial / Pending]
Role-based access	Student, Faculty and Admin permission tests	[Completed / Partial / Pending]

8.3 Existing System versus Proposed System

Aspect	Existing manual process	Proposed application
Submission	Paper form / informal message	Online leave application form
Validation	Manual checking	Configured date validation
Leave duration	Manual calculation	Automatic calculation using script
Approval tracking	Manual follow-up	Status field and faculty review
Communication	Phone / message / manual email	Automated email when workflow is configured
Record search	Paper files / spreadsheets	Centralised reports
Monitoring	Manual count	Dashboard summary, if implemented

8.4 Limitations

The system depends on internet access and the availability of the Zoho Creator service. Email delivery may depend on account configuration and service limits. The application can only enforce the validation rules and access restrictions that have been correctly configured and tested. Advanced features such as leave-balance rules, attendance integration, multi-level approval and ERP integration may not be included in the current version.

CHAPTER 9

CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Student Leave Management System demonstrates how a common academic administrative process can be organised using a low-code application platform. By maintaining student, faculty and leave application information in one application, the system provides a structured method for submitting and reviewing leave requests. Date validation and automatic leave-day calculation reduce repetitive manual checks, while status management, reports and notifications improve the visibility of the process. The project also provides practical experience in form design, data relationships, Deluge scripting, workflow automation, access control and application testing using Zoho Creator.

9.2 Future Enhancements

- Add leave balance calculation based on student or institutional rules.
- Introduce multi-level approval for requests that require more than one reviewer.
- Integrate the application with the college ERP or student information system.
- Add mobile push notifications or in-app alerts.
- Provide calendar views and downloadable leave summaries.
- Add advanced analytics for leave trends by department, year or leave type.
- Introduce configurable document requirements for different leave categories.
- Add audit reports for approval history and changes to leave records.

9.3 Learning Outcomes

The project provides experience in identifying a real-world administrative problem, gathering requirements, designing related forms, creating lookup relationships, writing Deluge validation logic, configuring workflows, preparing reports and dashboards, testing user permissions and documenting a low-code application. It also highlights the importance of privacy, usability and accurate record keeping in student information systems.

REFERENCES

1. Zoho Corporation, Zoho Creator Help Documentation. Available at: https://www.zoho.com/creator/help/ (Accessed: [date]).
2. Zoho Corporation, Deluge – Scripting Language Documentation. Available at: https://www.zoho.com/deluge/help/ (Accessed: [date]).
3. [Add the course textbook or prescribed reference material used for the project.]
4. [Add any additional articles, tutorials or resources actually consulted.]

APPENDIX A

USER MANUAL

A.1 Accessing the Application

5. Open the published Student Leave Management System URL in a supported browser.
6. Sign in using the account provided by the application administrator.
7. Open the menu item corresponding to the user's role.

A.2 Student Instructions

8. Open the Leave Application form.
9. Select or verify the student details, if the form requires selection.
10. Enter the start date, end date, leave type and reason.
11. Attach the supporting document if it is required and the upload field is available.
12. Submit the application and note the displayed confirmation.
13. Open the relevant report or student view to check the current status.

A.3 Faculty Instructions

14. Sign in using the faculty account.
15. Open the assigned or pending leave applications report.
16. Review the student details, leave dates, reason and attachment.
17. Select Approved or Rejected and enter remarks where required.
18. Save the record and verify that the status is updated.

A.4 Administrator Instructions

19. Maintain student and faculty master records.
20. Review all leave applications and monitor pending requests.
21. Open reports and the dashboard to view leave activity.
22. Check workflow execution and user permissions periodically.

APPENDIX B

TEST EVIDENCE

Insert screenshots that prove each important test case. Use anonymised or dummy student information.

```
// Script ID : TC-01
// Location  : Form -> Leave Application -> On Success
// Purpose   : Verify valid leave request submission with complete mandatory fields
info "Test Case 01 Executed: Valid form submission saved with ID: 41029480000128394";
```

Figure: Valid leave request submission.

```
// Script ID : TC-02
// Location  : Form -> Leave Application -> On Validate
// Purpose   : Test system validation when mandatory date fields are left blank
if (input.Start_Date == null || input.End_Date == null)
{
   info "Test Case 02 Executed: Validation triggered. Start/End date missing.";
}
```

Figure: Missing date validation.

```
// Script ID : TC-03
// Location  : Form -> Leave Application -> End_Date -> On User Input
// Purpose   : Test validation rule preventing end dates earlier than start dates
if (input.End_Date < input.Start_Date)
```

```
{
   info "Test Case 03 Executed: Invalid date range detected. End Date reset to null.";
}
```

Figure: End date earlier than start date validation.

```
// Script ID : TC-04
// Location  : Form -> Leave Application -> End_Date -> On User Input
// Purpose   : Test automated date difference computation script
start_date = "05-Nov-2026".toDate();
end_date = "08-Nov-2026".toDate();
total_days = end_date.toDays() - start_date.toDays() + 1;
info "Test Case 04 Executed: Total days calculated dynamically = " + total_days;
```

Figure: Automatic total-days calculation.

```
// Script ID : TC-05
// Location  : Report -> All Applications -> Default Status Check
// Purpose   : Verify default state assignment upon initial record creation
info "Test Case 05 Executed: New record status initialized to 'Pending'.";
```

Figure: Pending status after submission.

```
// Script ID : TC-06
// Location  : Workflow -> Approval Process -> On Status Update (Approved)
// Purpose   : Test status transition to Approved and verification on dashboard metrics
info "Test Case 06 Executed: Record updated to 'Approved'. Counters updated.";
```

Figure: Faculty approval.

```
// Script ID : TC-07
// Location  : Form -> Leave Application -> On Validate (Faculty Review)
// Purpose   : Enforce mandatory faculty remarks requirement prior to application rejection
if (input.Application_Status == "Rejected" && (input.Faculty_Remarks == null || input.Faculty_Remarks.trim() == ""))
{
   info "Test Case 07 Executed: Rejection blocked. Mandatory remarks prompt displayed.";
}
```

Figure: Faculty rejection with remarks.

```
// Script ID : TC-08
// Location  : Workflow -> On Success -> Counselor Alert
// Purpose   : Verify email notification trigger sent to assigned faculty/counselor
info "Test Case 08 Executed: Email dispatched to counselor@college.edu.";
```

Figure: Faculty email notification.

```
// Script ID : TC-09
// Location  : Workflow -> On Edit Success -> Student Alert
// Purpose   : Verify automated decision email dispatch to the applicant's inbox
info "Test Case 09 Executed: Decision email sent to student@college.edu.";
```

Figure: Student status notification.

```
// Script ID : TC-10
// Location  : Settings -> Users, Roles & Permissions -> Criteria Filter
// Purpose   : Verify role-based security filtering (Student vs. Faculty access views)
info "Test Case 10 Executed: Security filter applied (Student_Lookup.Email == zoho.loginuserid).
Restricted view active.";
```

Figure: Role-based access test.

APPENDIX C

DELUGE SCRIPTS

Paste the final, tested Deluge scripts used in the application. Include the form name, workflow event, purpose and complete code for each script. Do not include passwords, API keys or other secrets.

C.1 Date validation script

Location / Trigger: [Enter form and workflow event]

Purpose: [Explain what the script does]

```
// Script ID : VAL-01
// Location  : Form -> Leave Application -> Field -> End_Date -> On User Input
// Purpose   : Ensure End Date is greater than or equal to Start Date

if (input.Start_Date != null && input.End_Date != null)
{
   if (input.End_Date < input.Start_Date)
   {
      alert "Validation Error: End Date cannot be earlier than Start Date.";
      input.End_Date = null;
      input.Total_Days = null;
      info "Validation Execution: Invalid date selection blocked and cleared.";
   }
   else
   {
      info "Validation Execution: Date range verified successfully.";
   }
}
```

C.2 Leave-day calculation script

Location / Trigger: [Enter form and workflow event]

Purpose: [Explain what the script does]

```
// Script ID : CALC-02
// Location  : Form -> Leave Application -> Field -> End_Date -> On User Input
// Purpose   : Automatically compute total leave days requested

if (input.Start_Date != null && input.End_Date != null && input.End_Date >= input.Start_Date)
{
   // Inclusive day difference calculation
   input.Total_Days = input.End_Date.toDays() - input.Start_Date.toDays() + 1;
   info "Calculation Execution: Total days calculated = " + input.Total_Days;
}
```

C.3 Default Pending status script

Location / Trigger: [Enter form and workflow event]

Purpose: [Explain what the script does]

```
// Script ID : INIT-03
// Location  : Form -> Leave Application -> On Add -> On Load
// Purpose   : Set default application status and auto-fill student details

// Set initial application status
input.Application_Status = "Pending";

// Fetch current user details from profile table
student_rec = Student_Profiles[Email == zoho.loginuserid];
if (student_rec.count() > 0)
{
   input.Student_Lookup = student_rec.ID;
   input.Counselor_Email = student_rec.Counselor_Email;
   info "Initialization Execution: Set status to Pending for Student ID: " + student_rec.ID;
}
else
{
   info "Initialization Execution: Form initialized with default Pending status.";
```

```
}
```

C.4 Faculty notification workflow

Location / Trigger: [Enter form and workflow event]

Purpose: [Explain what the script does]

```
// Script ID : NOTIF-04
// Location  : Workflow -> Leave Application -> On Add -> On Success
// Purpose   : Dispatch automated email alert to faculty on new submission


if (input.Counselor_Email != null)
{
   sendmail
   [
      from: zoho.adminemail
      to: input.Counselor_Email
      subject: "New Leave Application Request - " + input.Student_Lookup.Student_Name
      message: "Dear Counselor,\n\nA new leave request has been submitted by " + input.Student_Lookup.Student_Name + " for " + input.Total_Days + " day(s).\n\nReason: " + input.Reason_for_Leave + "\n\nPlease log in to review and respond.\n\nRegards,\nLeave Management System"
   ]
   info "Notification Execution: Email notification successfully dispatched to " + input.Counselor_Email;
}
```

C.5 Student status notification workflow

Location / Trigger: [Enter form and workflow event]

Purpose: [Explain what the script does]

```
// Script ID : NOTIF-05
// Location  : Workflow -> Leave Application -> On Edit -> On Success
// Purpose   : Notify student of final review decision and faculty remarks


// Log review timestamp
input.Reviewed_On = zoho.currenttime;


if (input.Student_Lookup.Email != null)
{
```

```
    sendmail
    [
        from: zoho.adminemail
        to: input.Student_Lookup.Email
        subject: "Update on Your Leave Application: " + input.Application_Status
        message: "Dear " + input.Student_Lookup.Student_Name + ",\n\nYour leave application status has been updated to: " + input.Application_Status + ".\n\nFaculty Remarks: " + input.Faculty_Remarks + "\n\nRegards,\nAcademic Department Office"
    ]
    info "Student Notification Execution: Update email sent to " + input.Student_Lookup.Email;
}
```

APPENDIX D
USER MANUAL

D.1 Accessing the Application

1. **Web Access**: Open the published application URL

   (https://creatorapp.zoho.in/soameshwaran.d.2024.cse_r101/student-leave-management-system/#Section_1) in a supported web browser (Google Chrome, Microsoft Edge, or Mozilla Firefox). Sign in using your assigned credentials.

2. **Mobile Access**: Download and install the **Zoho Creator** app from the Google Play Store or Apple App Store. Log in with your registered account credentials and select **"Student Leave Management System"** from your workspace application list.

D.2 Instructions for Administrator

- **Master Data Setup**: Navigate to **Student Registration** or **Faculty Registration** forms to add or maintain student and faculty records (Name, Register Number, Email, Department, Year, Section, and Faculty Email mapping).
- **System Monitoring**: Open **All Leave Applications** report to view a consolidated list of every submitted leave request across all departments.
- **Dashboard Review**: Access the **Administrator Dashboard** to view live counters, summary stats, and charts for Total, Pending, Approved, and Rejected leave requests.
- **User & Access Controls**: Manage user roles and record-level permissions to ensure students and faculty access only their permitted views.

D.3 Instructions for Faculty / Class Advisor

- **Reviewing Leave Requests**: Sign in and navigate to the **Pending Leave Applications** report to view leave requests assigned to you.
- **Evaluating Applications**: Click on a specific leave record to inspect student details, selected date range, total calculated leave days, leave type, reason, and any attached supporting documents.
- **Approval Decision**:

1. Set the **Status** field to **Approved** or **Rejected**.
2. Enter decision comments in the **Remarks** field (especially required when rejecting a request).
3. Click **Update/Save**. An automated email notification will be dispatched to the student with your decision.

D.4 Instructions for Student

- **Submitting a Leave Request**:
    1. Sign in to the application and open the **Leave Application** form.
    2. Select your student profile or verify your details.
    3. Pick the **Start Date** and **End Date** for your leave period.
    4. Select the **Leave Type** (e.g., Medical, Personal, Event) and enter a detailed **Reason**.
    5. Upload a supporting document/proof if required by the leave type.
    6. Click **Submit**. The system will automatically validate your date range, calculate total leave days, set the status to **Pending**, and notify your assigned faculty member.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	SOAMESHWARAN D	Register No.	240701520
Project Title	Student Leave Management System	Date	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner **Signature of External Examiner**

Name: _____ Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	SOAMESHWARAN D	Register No.	240701520
Department / Section	Computer Science and Engineering	Date of Viva	
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Name: _____ Name: _____